
BENJAMIN DOURTHE, PhD

AI Strategist & Machine Learning Leader | Digital Health & MedTech | Team & Program Builder

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SUMMARY

AI and data science leader with 10+ years across digital health, MedTech R&D, and regulated environments. Built and scaled machine learning, computer vision, and signal-processing systems from prototype to product, including first-in-class wearable, implantable and imaging solutions. Known for fast, robust iteration, strong engineering discipline (automation, testing, reproducibility), and cross-functional leadership across software, engineering, clinical, and research teams.

CORE COMPETENCIES

Digital health & MedTech AI: biomedical sensing, medical imaging, digital biomarkers, wearables, implantables.

Agentic AI: platform rollout, prompt engineering, governance, security/compliance reviews.

Machine learning & deep learning: classification/regression, CNNs/RNNs, generative models, segmentation.

Computer vision & imaging: reconstruction, super-resolution (SRGAN), 2D/3D pipelines, quality optimization.

Time-series & signal processing: feature engineering, denoising, model robustness, generalization.

Production-grade Python: data pipelines, experiment tracking, code quality, automation, Git.

Leadership: hiring/interviewing, mentoring, roadmap delivery, stakeholder communication.

TECHNICAL SKILLS

Programming Languages: Python (primary), MATLAB, R, SQL, C, C++, Java, JavaScript, HTML/CSS.

Machine Learning & Deep Learning: scikit-learn, PyTorch, TensorFlow, hyperparameter optimization, model evaluation/validation, robustness testing, explainability.

Generative AI & Agentic Systems: Prompt and context engineering (advanced system prompts, reusable prompt frameworks), RAG design/build, org/team/project-specific skills; tools: Claude Code, OpenAI Codex, Gemini Enterprise, Cursor, Windsurf, GitHub Copilot, Deep Research.

Data, Signals & Imaging: NumPy, Pandas, SciPy, OpenCV, time-series analysis, feature engineering, multimodal data fusion, clinical-grade analytics.

DevOps, CI/CD & Engineering Practices: Git, testing, automation, documentation, reproducible research; CI/CD for small teams; repo architecture/branching strategies; data management for security/privacy/compliance; data & AI policy development.

Backend/Frontend & Full-Stack Foundations: APIs, backend and frontend frameworks, internal tooling and workflow automation for ops-heavy teams.

Domains: Digital health, MedTech, medical imaging, biomechanics, cardiac/pulmonary monitoring, wearables and sensor analytics; startup and early-stage product development.

PROFESSIONAL EXPERIENCE

STAFF MACHINE LEARNING ENGINEER, Shifamed / Adona Medical - Los Gatos, CA (2024 - Present)

MedTech company developing implantable cardiac monitoring systems.

- Lead enterprise rollout of Generative and Agentic AI platforms (strategy, enablement, and governance), including security/compliance reviews and internal controls to protect sensitive data and meet regulatory requirements.
- Design and deliver cross-functional Generative and Agentic AI training, handbooks, and office hours; internal survey measured ~20% average productivity lift across engineering and non-engineering teams, increasing throughput, creativity, and work quality while driving meaningful cost savings.
- Develop automated and agentic data-processing and analytics pipelines, as well as custom project- and team-specific agentic applications across Software, EE, and R&D to accelerate iteration, improve robustness, and provide near real-time operational insights.

- Engineer and validate signal processing and machine learning algorithms for complex cardiac pressure data to support next-generation clinical tools and data infrastructure.
- Founded and leading the AI & Data Science operating model across algorithm development, data standardization & governance, AI tool enablement/optimization, and custom agentic tools tailored to internal workflows.
- Built and maintained an internal agentic AI engineering toolkit with 277 production-ready prompt and context engineering templates across 7 languages, enabling consistent, high-quality AI-assisted development across teams.
- Authored compliance-oriented templates spanning SOC 2, ISO 27001, GDPR/CCPA, NIST AI RMF and agent governance patterns (lifecycle/risk/security/observability) to support regulated development practices.

SENIOR MACHINE LEARNING ENGINEER, Nuralogix Corporation - Toronto, Canada (2022 - 2023)

Digital health company using computer vision for video-based physiological monitoring.

- Led end-to-end design and optimization of signal-processing + ML pipelines, improving video data quality by 25% and biometric prediction performance by 12% (internal benchmark).
- Engineered 100+ high-value features for high-dimensional data representation, enabling stronger predictive performance and faster hypothesis testing.
- Introduced robustness-focused evaluation and model generalization strategies for real-world, multi-modal health data.

CHIEF DATA OFFICER, Tidal Medical - Toronto, Canada (2020 - 2022)

Startup developing first-in-class non-invasive wearable for real-time cardiopulmonary monitoring.

- Defined the data/AI architecture for a first-in-class non-invasive wearable for real-time pulmonary artery pressure monitoring, achieving 97% accuracy and enabling heart-failure management use cases.
- Pioneered deep learning approaches for medical imaging enhancement, including SRGAN-based super-resolution to improve diagnostic image quality.
- Improved software/algorithm efficiency by ~90% via redesign of signal-processing and computer-vision pipelines, enabling faster prototyping and more robust deployments.
- Built and led multidisciplinary teams; drove IP strategy and execution (1 PCT + 4 US provisional filings) and secured >\$2M in non-dilutive funding across national programs and grants.

MACHINE LEARNING ENGINEER (Postdoctoral), University of British Columbia - Vancouver, Canada (2020 - 2022)

- Designed and deployed deep learning models for medical image segmentation (e.g., multi-scale architectures) improving accuracy and workflow efficiency for clinical research teams.
- Collaborated with corporate partners (Medtronic) and clinical stakeholders to deliver solutions suitable for real-world medical environments.
- Published peer-reviewed work in Spine (2022) on automated segmentation of spinal muscles from upright open MRI.

COMPUTER VISION ENGINEER (Postdoctoral), Sunnybrook Research Institute - Toronto, Canada (2018 - 2020)

- Developed marker-less human motion analysis using RNN/LSTM and computer-vision pipelines, improving tracking accuracy by 35% for non-invasive monitoring and rehabilitation applications.
- Integrated multiple sensing modalities to advance real-time biomechanical data analysis in research and clinical settings.

RESEARCH ENGINEER (PhD), KU Leuven - Belgium (2014 - 2017)

- Developed patient-specific computational models combining medical imaging (CT) with biomechanical simulation for surgical planning applications.
- Built image processing algorithms for 3D morphological feature extraction and stress mapping from clinical imaging data.

EDUCATION

- **PhD, Biomedical Engineering** - KU Leuven, Belgium (2014 - 2017)
- **MSc, Science, Technology and Health** - University of Lorraine, France (2012 - 2013)
- **MSc, Mechanical Engineering** - National Engineering School of Metz (ENIM), France (2008 - 2013)
- *Visiting Scholar* - University of British Columbia, Canada (Fall 2016); University of Southern California, USA (Fall 2015)

SELECTED PUBLICATIONS & INDEPENDENT RESEARCH

- **"The Future of Intelligence" (2025)** - 47-page independent essay analyzing the limitations of current AI architectures (LLMs, Transformers) and proposing research directions toward systems with genuine understanding. Includes mathematical foundations of attention mechanisms, predictive coding, and world models. <https://doi.org/10.5281/zenodo.18177869>
- **8+ peer-reviewed publications** in computational modeling, medical imaging, and applied machine learning.
- **5 patents/patent applications** in medical AI and sensing systems.